

Sabia: AI Research Insights

Technical Report for the REST API and Integrated GUI

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Technical Report for the REST API and Integrated GUI

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Source inspected: `blue-demon:/home/iarroyof/sabia/ai-research-insights`

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1 Executive Summary

Sabia: AI Research Insights is a containerized biomedical research platform for searching scientific evidence, selecting relevant sentences/triplets, generating context-grounded summaries and chat answers, and visualizing extracted biomedical knowledge as an interactive graph. The system combines a Streamlit GUI, a FastAPI REST API, tenant-aware middleware, hybrid search, object storage, relational annotation storage, graph storage, GPU-backed LLM inference, relation extraction, and background workers.

The current deployed API surface is defined by `services/api/app/main.py`. It mounts the `search`, `chat`, `triplets`, `papers`, `annotations`, and CSV ingest routers. Several additional router files exist in the repository but are not mounted; they are documented separately because source presence alone does not make them reachable through the running FastAPI app. The dataset and lung-cancer knowledge-base construction context is aligned with the scalable lung cancer knowledge-base work reported by Martínez Cisneros et al. (2025), while the semantic reasoning and attention-based modeling context is aligned with Balderas-Martínez et al. (2025).

2 1. Project Overview and Architecture

2.1 Core Problem

Biomedical literature review requires repeated movement between papers, snippets, relation extraction output, and expert interpretation. The project addresses this by combining:

- search over indexed scientific sentences and extracted triplets;
- user selection of relevant evidence;
- LLM-assisted chat and conditioned summarization grounded in selected context;
- entity/relation triplet extraction and filtering;
- interactive knowledge graph exploration;
- annotation capture for later review, curation, or model-improvement workflows.

The software is not just a search UI. It is an evidence workbench where the REST API exposes the retrieval, ingestion, summarization, graph, and annotation capabilities, while the GUI orchestrates those capabilities into a researcher-facing workflow. Its data and knowledge-base workflow should be read as an implementation-oriented continuation of scalable biomedical knowledge-base construction methods for lung cancer literature (Martínez Cisneros et al., 2025).

2.2 Logical Architecture

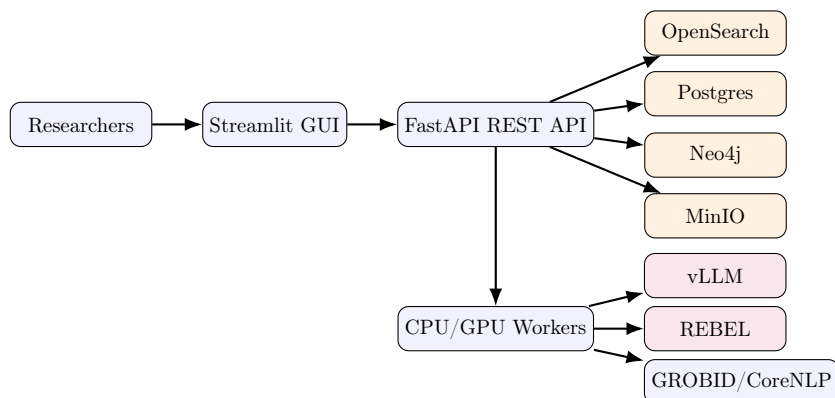


Figure 1: Logical architecture of the Sabia platform.

Mermaid source: [docs/diagrams/logical-architecture.mmd](#)

2.3 Physical Deployment Architecture

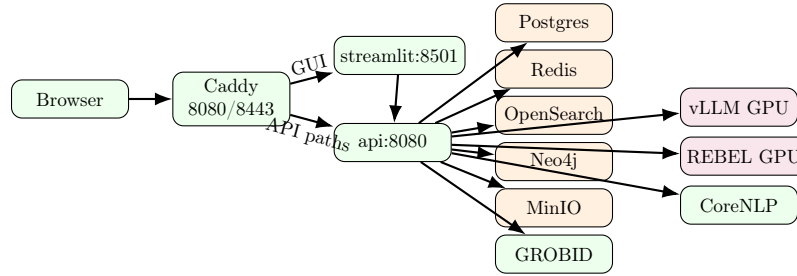


Figure 2: Physical Docker Compose deployment and service communication.

Mermaid source: [docs/diagrams/physical-deployment.mmd](#)

2.4 Main Runtime Components

- **Caddy** listens on ports 8080 and 8443. It reverse-proxies API paths such as `/search*`, `/chat*`, `/papers*`, `/triplets*`, `/annotations*`, and `/health` to the FastAPI service. All other paths are routed to Streamlit.
- **FastAPI API** runs `uvicorn app.main:app --host 0.0.0.0 --port 8080` and exposes the REST API.
- **Streamlit GUI** provides the researcher workflow and calls the API using `httpx`.
- **Postgres** stores relational application data such as annotations and uses a tenant context intended for RLS policies.
- **OpenSearch** stores searchable sentence/triplet indexes and supports hybrid lexical/vector retrieval.
- **Neo4j** stores graph representations for entities and relations.
- **MinIO** stores uploaded PDF/TXT objects.
- **Redis/Celery workers** support asynchronous ingestion and processing work.
- **GROBID** extracts structured metadata from scientific PDFs.
- **vLLM** serves an OpenAI-compatible local LLM endpoint using `HuggingFaceH4/zephyr-7b-beta`.
- **REBEL** is the primary relation extraction service. `CoreNLP/OpenIE` is configured as a fallback path.

2.5 Technology Stack

Layer	Technologies Observed
API framework	Python, FastAPI 0.112.0, Uvicorn 0.30.5, Pydantic 2.8.2, Pydantic Settings
GUI	Streamlit 1.38.0, <code>httpx</code> , Streamlit session state, Streamlit secrets
Search	OpenSearch 2.13.0, BM25/vector hybrid settings, sentence-transformers optional embeddings
Data storage	Postgres 16, MinIO, Neo4j 5.23.0-community, Redis
Background processing	Celery 5.4.0, CPU/GPU worker containers
NLP/AI	vLLM OpenAI-compatible server, Zephyr 7B, REBEL, Stanford CoreNLP, GROBID, transformers, torch, spaCy/scispaCy
Proxy/deployment	Docker Compose, Caddy, NVIDIA runtime for GPU containers
API documentation style	FastAPI/OpenAPI-ready route definitions, supplemented here by source-derived technical documentation

The semantic modeling layer combines indexed triplets, vector search, entity/relation representations, and context-conditioned LLM prompting. This design is consistent with prior work on semantic reasoning with standard attention-based models in chronic disease literature, where attention-based representations are used to support semantic inference over biomedical text (Balderas-Martínez et al., 2025).

3 2. REST API Specification

3.1 Base URLs

- Internal Docker network API URL: <http://api:8080>
- Host-published API URL in Compose: <http://localhost:18081>

- Caddy public API paths: <http://localhost:8080/<api-path>>
- Streamlit public path: <http://localhost:8080/> through Caddy fallback routing

3.2 Required Headers

Header	Required	Purpose
X-API-Key	Required for all non-public endpoints	Compared against <code>API_KEY</code> environment variable or <code>security.api_key</code> from config.
X-Tenant-Id	Required for all normal endpoints except <code>/health</code> ; optional/query fallback for <code>/triplets/graph/view</code>	Sets <code>request.state.tenant_id</code> ; used by storage/search/annotation operations.
Content-Type: application/json	Required for JSON POST/PUT requests	Tells FastAPI to parse JSON bodies.
x-user-id	Optional for annotation endpoints	Stored as annotation <code>user_id</code> when provided.

Public paths in current middleware: `/health` and `/triplets/graph/view*`.

3.3 Deployed Endpoint Inventory

Method	URI	Tag	Auth	Purpose
GET	<code>/health</code>	Health	Public	Returns <code>{'status':'ok'}</code> .
POST	<code>/search/</code>	Search	API key + tenant	Hybrid search over papers/triplets/chats target model. Returns items alias for hits.
POST	<code>/chat/</code>	Chat	API key + tenant	Streams server-sent events with token, citations, and final messages.
POST	<code>/papers/upload</code>	Papers	API key + tenant	Multipart PDF/TXT upload to MinIO; enqueues URI ingestion when task helper is wired.
POST	<code>/papers/link</code>	Papers	API key + tenant	Accepts article/direct links and enqueues link ingestion when task helper is wired.
POST	<code>/papers/ncbi</code>	Papers	API key + tenant	Accepts PMID list and enqueues NCBI ingestion when task helper is wired.
POST	<code>/papers/summarize_conditioned</code>	Papers	API key + tenant	RAG/LLM summary over selected items; returns paragraphs and support sentences.
POST	<code>/triplets/csv</code>	Triplets	API key + tenant	Uploads CSV triplets; validates entity probabilities and confidence thresholds.
POST	<code>/triplets/ingest_csv</code>	Triplets	API key + tenant	Uploads CSV through <code>ingest_csv_triplets</code> helper.
POST	<code>/triplets/suggest</code>	Triplets	API key + tenant	Runs relation extraction on supplied sentences.
POST	<code>/triplets/graph/build</code>	Triplets	API key + tenant	Builds graph candidate triple IDs from selected sentences.
GET	<code>/triplets/graph</code>	Triplets	API key + tenant	Returns graph JSON for comma-separated triple IDs.
GET	<code>/triplets/graph/view</code>	Graph	Public API key bypass; tenant header/query	Returns browser-friendly vis-network HTML graph viewer.
GET	<code>/triplets/search</code>	Triplets	API key + tenant	Searches triplet index by <code>q</code> , <code>confidence_min</code> , <code>entity_type</code> , and <code>pmid</code> .
POST	<code>/triplets/classify</code>	Classification	API key + tenant	Classifies entities with the NER/zero-shot classifier.
POST	<code>/triplets/sync</code>	Sync	API key + tenant	Manually syncs OpenSearch triplets into Neo4j.
POST	<code>/annotations</code>	Annotations	API key + tenant	Creates one annotation with optional idempotency key.
POST	<code>/annotations/batch</code>	Annotations	API key + tenant	Creates up to 500 annotations.
GET	<code>/annotations</code>	Annotations	API key + tenant	Lists annotations with filters and cursor pagination.
PUT	<code>/annotations/{annotation_id}</code>	Annotations	API key + tenant	Updates kind, payload, page, or bbox.
DELETE	<code>/annotations/{annotation_id}</code>	Annotations	API key + tenant	Deletes annotation for current tenant.

GET	/annotations/export	Annotations	API key + tenant	Streams NDJSON annotations for downstream training/RLHF workflows.
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3.4 Source Routers Not Currently Mounted

These files exist under `services/api/app/routers`, but `services/api/app/main.py` does not include them with `app.include_router(...)`.

Method	URI	Source	Status
POST	/chat/history/search	chat_history.py	Defined as POST but not mounted; Streamlit currently calls GET, so the GUI history search will not work unless mounted and aligned.
GET	/chat/session/{session_id}/messages	chat_sessions.py	Defined but not mounted.
POST	/export/triples.csv	export.py	Defined but not mounted.
POST	/jobs/backfill/probabilities	jobs.py	Defined but not mounted.
POST	/predict/probabilities	probabilities.py	Defined but not mounted.

3.5 Request and Response Models

Model	Fields
SearchRequest	query: str, target: papers triplets chats all = all, filters: object = {}, k: 1..200 = 20
SearchResponse/SearchHit	items: list of hits with paper_id, title, pmid, pmcid, page, sent_id, score, text, subject, relation, object, confidence.
ChatRequest	message: str, items: ChatItem[], session_id?: str, options: {allow_extra_retrieval, token_budget, confidence_min}.
ConditionedSummaryRequest	message: str, items: ConditionedSummaryItem[], options: object.
ConditionedSummaryResponse	paragraphs: list of {text, support[]} where support carries sentence provenance and SVO fields.
UploadResponse	job_id, num_files, objects stored as s3://bucket/tenant/uploads/....
AnnotationCreate	paper_id, sent_id?, kind: highlight note like unlike, payload, page?, bbox?, idempotency_key?.
AnnotationOut	id, paper_id, sent_id, kind, payload, page, bbox, user_id, created_at.
Triplet graph build response	triple_ids, count, and debug metadata with selected item count, paper IDs, query terms, thresholds, and top_k.
Classify response	results from NER classifier and models_available.

3.6 Request and response examples

3.6.1 Health

```
curl http://localhost:8080/health
```

```
{"status": "ok"}
```

3.6.2 Search

```
curl -X POST http://localhost:8080/search/ \
-H "X-API-Key: $API_KEY" \
-H "X-Tenant-Id: default" \
-H "Content-Type: application/json" \
-d '{"query":"immunotherapy lung carcinoma","target":"all","filters":{"confidence_min":0.6},"k":3}'
```

```
{
  "items": [
    {
      "paper_id": "PMC123456.txt",
      "pmid": "12345678",
      "pmcid": "PMC123456",
      "sent_id": "sent-0001",
      "score": 12.84,
      "text": "PD-1 blockade improved response in lung carcinoma cohorts.",
      "subject": "PD-1 blockade",
      "relation": "improved",
      "object": "response",
      "confidence": 0.87
    }
  ]
}
```

Common errors:

```
{"detail": "Invalid X-API-Key"}
```

```
{"detail": "Search failed: RuntimeError: <backend message>"}
```

3.6.3 Chat with pinned context

The chat endpoint returns `text/event-stream`; clients should parse each `data:` line as JSON.

```
curl -N -X POST http://localhost:8080/chat/ \
-H "X-API-Key: $API_KEY" \
-H "X-Tenant-Id: default" \
-H "Content-Type: application/json" \
-d '{"message": "Summarize the evidence.", "items": [{"paper_id": "PMC123456.txt", "sent_id": "sent-0001", "text": "PD-1 blockade improved
↪ response."}], "options": {"allow_extra_retrieval": false}}'
```

```
data: {"type": "token", "data": "PD-1 blockade..."}
data: {"type": "citations", "data": {"snippets": [{"paper_id": "PMC123456.txt", "sent_id": "sent-0001"}]}}
data: {"type": "final", "data": {"done": true, "session_id": "6fd7e1d1-1c9b-4b5d-9a10-8ac5e9f27e33"}}
```

3.6.4 Upload papers

```
curl -X POST http://localhost:8080/papers/upload \
-H "X-API-Key: $API_KEY" \
-H "X-Tenant-Id: default" \
-F "files=@article.pdf"
```

```
{
  "job_id": "8a6e0f31-7e8a-4e83-b39e-bb971c2b0793",
  "num_files": 1,
  "objects": ["s3://papers/default/uploads/uuid_article.pdf"]
}
```

If the ingestion task helper is not wired:

```
{"detail": "Ingestion task not wired. Implement app.tasks.ingest.enqueue_ingest_uris(tenant, uris) to process uploaded objects."}
```

3.6.5 Link and PMID ingestion

```
[
  {"url": "https://pubmed.ncbi.nlm.nih.gov/12345678/", "metadata": {"topic": "immunotherapy"}}
]

{"job_id": "link-job-id", "num_links": 1}

{"pmids": ["12345678", "23456789"]}

{"job_id": "pmid-job-id", "num_pmids": 2}
```

3.6.6 Conditioned summary

```
{
  "message": "What are the main immunotherapy strategies for lung carcinoma?",
  "items": [
    {
      "paper_id": "PMC123456.txt",
      "sent_id": "sent-0001",
      "text": "PD-1 blockade improved response in lung carcinoma.",
      "subject": "PD-1 blockade",
      "relation": "improved",
      "object": "response"
    }
  ],
  "options": {"max_paragraphs": 3, "confidence_min": 0.5}
}

{
  "paragraphs": [
    {
      "text": "The selected evidence emphasizes immune checkpoint inhibition...",
      "support": [
        {
          "paper_id": "PMC123456.txt",
          "sent_id": "sent-0001",
          "text": "PD-1 blockade improved response in lung carcinoma.",
          "subject": "PD-1 blockade",
          "predicate": "improved",
          "object": "response",
          "pmid": "12345678",
          "pmcid": "PMC123456",
          "page": 4
        }
      ]
    }
  ]
}
```

3.6.7 CSV triplet upload

```
curl -X POST http://localhost:8080/triplets/csv \
-H "X-API-Key: $API_KEY" \
-H "X-Tenant-Id: default" \
-F "file=@triplets.csv"
```

```
{"accepted": 1200, "filtered": 83, "job_id": "csv:6c4f8040-7b33-4cf8-b7a4-d7143b646736"}
```

Error:

```
{"detail": "Invalid file type (expected .csv)"}
```

3.6.8 Suggest triplets

```
[
  "Pembrolizumab targets PD-1 in lung cancer therapy."
]
```

```
[
  {
    "subject": "Pembrolizumab",
    "predicate": "targets",
    "object": "PD-1",
    "confidence": 0.81
  }
]
```

3.6.9 Build graph from selected rows

```
curl -X POST "http://localhost:8080/triplets/graph/build?confidence_min=0.6&ebio_min=0.3&top_k=50" \
-H "X-API-Key: $API_KEY" \
-H "X-Tenant-Id: default" \
-H "Content-Type: application/json" \
-d '[{"paper_id": "PMC123456.txt", "sent_id": "sent-0001", "text": "PD-1 blockade improved response."}]'
```

```
{
  "triple_ids": ["triplet-1", "triplet-2"],
  "count": 2,
  "debug": {
    "mode": "semantic_search_filtered",
    "items_in": 1,
    "paper_ids": ["PMC123456.txt"],
    "query_terms": "PD-1 blockade improved response",
    "triples_found": 2,
    "confidence_min": 0.6,
    "ebio_min": 0.3,
    "top_k": 50
  }
}
```

3.6.10 Graph JSON and viewer

```
curl "http://localhost:8080/triplets/graph?triple_ids=triplet-1,triplet-2&confidence_min=0.6" \
-H "X-API-Key: $API_KEY" \
-H "X-Tenant-Id: default"

{
  "nodes": [{"id": "PD-1", "label": "PD-1", "type": "protein"}],
  "edges": [{"id": "triplet-1", "source": "Pembrolizumab", "target": "PD-1", "label": "targets", "confidence": 0.81}]
}
```

The HTML viewer is browser-facing and accepts the API key bypass in middleware:

```
http://localhost:8080/triplets/graph/view?triple_ids=triplet-1,triplet-2&confidence_min=0.6&tenant=default
```

3.6.11 Annotation CRUD

```
{
  "paper_id": "PMC123456.txt",
  "sent_id": "sent-0001",
  "kind": "note",
  "payload": {"text": "Strong evidence for checkpoint inhibition."},
  "page": 4,
  "bbox": {"x": 100, "y": 220, "width": 250, "height": 40},
  "idempotency_key": "client-generated-key"
}
```



```
{
  "id": "9df9f20c-6dd4-4e6d-aacf-4bbf8eab8b24",
  "paper_id": "PMC123456.txt",
  "sent_id": "sent-0001",
  "kind": "note",
  "payload": {"text": "Strong evidence for checkpoint inhibition."},
  "page": 4,
  "bbox": {"x": 100, "y": 220, "width": 250, "height": 40},
  "user_id": "researcher-1",
  "created_at": "2026-04-24T12:00:00Z"
}
```

List response:

```
{
  "items": [
    {
      "id": "9df9f20c-6dd4-4e6d-aacf-4bbf8eab8b24",
      "paper_id": "PMC123456.txt",
      "sent_id": "sent-0001",
      "kind": "note",
      "payload": {
        "text": "Strong evidence for checkpoint inhibition.",
        "page": 4,
        "bbox": {
          "x": 100,
          "y": 220,
          "width": 250,
          "height": 40
        }
      },
      "user_id": "researcher-1",
      "created_at": "2026-04-24T12:00:00Z"
    }
  ],
  "next_cursor": null
}
```

Delete response:

```
{
  "status": "ok",
  "deleted": 1
}
```

Common annotation errors:

```
{
  "detail": "Missing X-Tenant-Id"
}
{
  "detail": "Annotation not found"
}
{
  "detail": "Invalid cursor"
}
```

3.6.12 Entity classification

```
{
  "entities": ["PD-1", "lung carcinoma", "immune checkpoint inhibitor"]
}
{
  "results": [
    {
      "text": "PD-1",
      "entity_type": "biomedical",
      "scores": {
        "EBio": 0.91,
        "NGen": 0.06,
        "other": 0.03
      }
    }
  ],
  "models_available": ["facebook/bart-large-mnli"]
}
```

If models are unavailable:

```
{
  "detail": "NER models not loaded. Please install models first."
}
```

3.6.13 Manual Neo4j sync

```
curl -X POST "http://localhost:8080/triplets/sync?max_triplets=5000" \
  -H "X-API-Key: $API_KEY" \
  -H "X-Tenant-Id: default"

{
  "status": "success",
  "tenant": "default",
  "index": "triplets_default",
  "stats": {
    "nodes": 1200,
    "relationships": 980
  }
}
```

3.7 Authentication and Security

Authentication is API-key based. The middleware accepts the key in either:

- **X-API-Key:** `<value>` header, which is the primary method;
- **?api_key=<value>** query parameter, which is a fallback for browser-only flows.

The expected key is resolved from `API_KEY` first and then from `settings.security.api_key`. If `security.require_api_key` is false, key checking is bypassed. The configuration file also contains `enable_hmac` and `rate_limit_per_min`, but the inspected middleware does not implement HMAC verification or rate limiting yet.

Tenant isolation is handled by `TenantMiddleware`, which requires `X-Tenant-Id` for normal endpoints and stores it as `request.state.tenant_id`. The annotations router sets `app.tenant_id` in Postgres using `set_config(...)`, which is intended to align with row-level security policies.

Security implications:

- API keys are long-lived shared secrets; rotate them through `.env/deployment` secret management.
- Do not expose non-public endpoints without HTTPS in production.
- `/triplets/graph/view` is intentionally public in middleware; it should only expose non-sensitive graph IDs or be protected before external deployment.
- Query-string API keys are convenient for browsers but can leak through logs and history; prefer headers where possible.
- HMAC and rate limiting are configured but not enforced in the inspected middleware.

3.8 Error Handling and Troubleshooting

Code	Current Sources	Meaning	Troubleshooting
400	Tenant middleware, CSV upload, annotations, graph	Missing tenant, invalid cursor, invalid file type, empty update body, missing triple IDs/entities	Check X-Tenant-Id , query parameters, file extension, request body, and required IDs.
403	Security middleware	Missing or invalid API key	Confirm X-API-Key equals API_KEY in the API container environment.
404	Annotation update/delete	Annotation not found for current tenant	Verify annotation_id and tenant context.
422	FastAPI/Pydantic	Invalid schema, missing fields, invalid ranges/patterns	Check request body against the models above.
500	Search, annotations DB, Neo4j sync, graph viewer	Backend exception	Inspect API logs, upstream service availability, and tenant/index names.
501	Papers ingestion, triplet search	Optional helper/backend not wired	Implement the referenced task helper or enable the missing backend.
502	Paper upload to MinIO	Object storage failure	Check MinIO service, bucket, credentials, and network reachability.
503	Entity classifier	NER models not loaded	Install/load spaCy/scispaCy/zero-shot models and restart the API.

Recommended improvement: standardize error bodies into a stable shape such as `{"error": {"code": "...", "message": "...", "details": ...}}`, add request IDs, and document which errors are retryable.

4 3. GUI and Integration Design

4.1 GUI User Journey

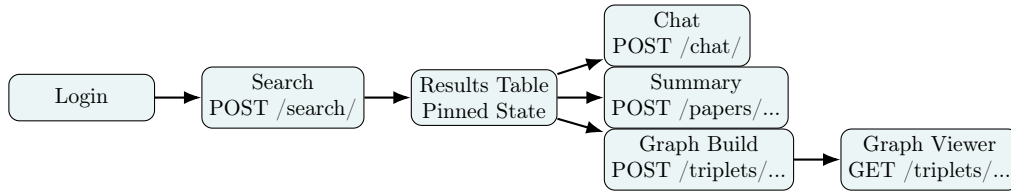


Figure 3: GUI user journey and API interaction flow.

Mermaid source: [docs/diagrams/gui-api-flow.mmd](#)

4.2 Component and API Data Source Map

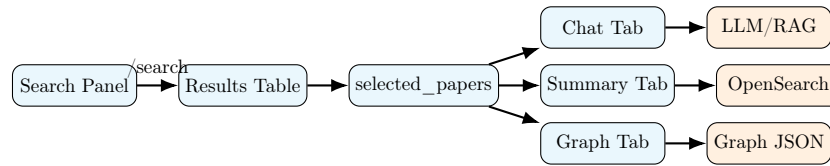


Figure 4: GUI components and their backend data sources.

Mermaid source: [docs/diagrams/component-map.mmd](#)

4.3 Search-to-Graph Sequence

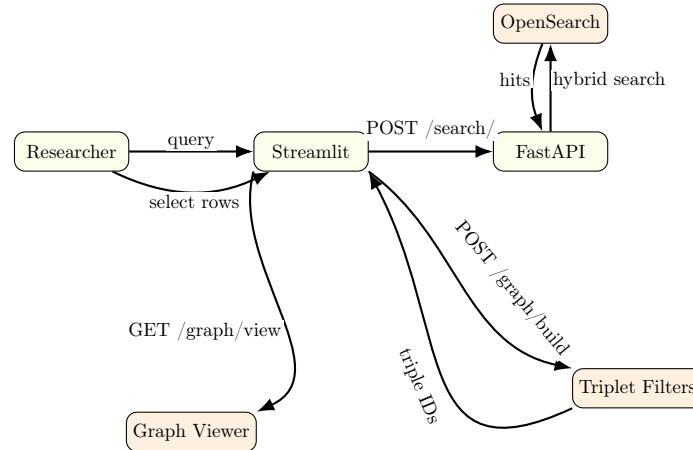


Figure 5: Search-to-graph sequence from query to interactive visualization.

4.4 Main Screens and Intended Flow

1. **Login screen:** Streamlit checks username/password against `st.secrets.passwords`. Successful login sets `st.session_state[`
= True.
2. **Search panel:** The user enters a biomedical query and Top K. The GUI sends `POST /search/` with `query`, `target`,
filters, and `k`.
3. **Results table:** Results are sorted descending by `confidence`, displayed with subject/relation/object/text/provenance
fields, and selected using checkbox cells in `st.data_editor`.
4. **Pinned context state:** Selected rows are normalized into `st.session_state["selected_papers"]`.
5. **Chat tab:** The selected items are sent to `POST /chat/`. The GUI streams SSE tokens and later displays citations.
6. **Conditioned summary tab:** The selected items plus a question are sent to `POST /papers/summarize_conditioned`.
7. **Triplets/Graph tab:** The selected items, confidence threshold, biomedical score threshold, and Top K are sent to
`POST /triplets/graph/build`. The returned `triple_ids` are stored in `st.session_state["graph_triple_ids"]`.
8. **Graph viewer:** The GUI builds a URL for `GET /triplets/graph/view?triple_ids=...&confidence_min=...&tenant=...`
which returns a vis-network HTML graph.

4.5 API-UI Mapping

UI Action	GUI State/Input	API Operation	API Output	GUI Effect
Login	username, password	No API call; Streamlit secrets check	Boolean auth state	Allows or stops app rendering.
Search	q, k, tenant, API key	<code>POST /search/</code>	items list	Populates <code>results</code> , clears previous selection.
Select rows	Edited data table	No API call	N/A	Updates <code>selected_papers</code> .
Start Chat	message, selected items, allow_extra_retrieval	<code>POST /chat/</code>	SSE token/citations/final events	Streams answer and citations.
Run Conditioned Summary	question, selected items	<code>POST /papers/summarize_conditioned</code>	paragraphs with support	Displays summary and supporting evidence.
Build graph	selected items, confidence_min, ebio_min, top_k	<code>POST /triplets/graph/build</code>	triple_ids, debug details	Stores graph IDs and shows debug metadata.
Open graph viewer	graph_triple_ids, tenant, confidence	<code>GET /triplets/graph/view</code>	HTML page	Opens interactive graph in a new tab.
Search history	hist_q	Intended history search	Currently mismatched	GUI calls <code>GET /chat/history/search</code> , but source defines unmounted <code>POST /chat/history/search</code> .

4.6 State Management

The Streamlit GUI uses `st.session_state` as its client-side state container:

- **authenticated:** login state.
- **results:** latest search result set.

- `selected_papers`: normalized selected rows used by chat, summary, and graph workflows.
- `history_hits`: intended chat-history search result state.
- `graph_triple_ids`: latest graph build output.

Sorting is performed client-side by confidence after search results arrive. Paging is not implemented in the GUI search result table; the API accepts `k`, and the Streamlit input limits it to 1-100. Filtering for graph construction is server-side through query parameters, while graph-view filtering is client-side inside the returned HTML page.

4.7 Integration Finding

The GUI contains a history-search feature, but it currently does not line up with the deployed API:

- GUI call: `GET /chat/history/search` with query parameters.
- Source router: `POST /chat/history/search` with a `ChatSearchReq` JSON body.
- Deployment: `chat_history.py` is not mounted in `main.py`.

To make this feature operational, mount the router in `main.py` and either change the GUI to POST JSON or add a GET-compatible endpoint.

4. Development and Maintenance

5.1 Setup and Installation

Prerequisites:

- Docker and Docker Compose.
- NVIDIA Container Toolkit for GPU services (`worker-gpu`, `llm`, `rebel-extractor`).
- A `.env` file providing at least `API_KEY`, `POSTGRES_USER`, `POSTGRES_PASSWORD`, `POSTGRES_DB`, `NEO4J_USER`, `NEO4J_PASSWORD`, `MINIO_ROOT_USER`, `MINIO_ROOT_PASSWORD`, and `LLM_MODEL_ID` as needed.
- Hugging Face model cache mounted at `/mnt/data/huggingface/hub` on the host, or an adjusted model volume path.

Typical startup:

```
cd /home/iarroyof/sabia/ai-research-insights
make build
make up
make ps
make logs
```

Manual Compose equivalent:

```
docker compose build
docker compose up -d
docker compose ps
docker compose logs -f api streamlit
```

Health check:

```
curl http://localhost:8080/health
```

Streamlit through Caddy:

```
http://localhost:8080/
```

Host-published API port from Compose:

```
http://localhost:18081/
```

OpenSearch host port:

```
http://localhost:19200/
```

Neo4j browser:

```
http://localhost:7474/
```

5.2 Configuration

The API reads `APP_CONFIG=/app/config/default.yaml`, expands environment variables, and validates the config using Pydantic settings. Important sections:

- **security**: API key requirement, configured key, HMAC flag, configured rate limit.
- **multi_tenancy**: RLS strategy, default tenant, enforcement flag.
- **postgres**: DSN assembled from environment variables.
- **opensearch**: endpoint, vector dimension, BM25/vector retrieval settings, fusion weight.
- **neo4j**: Bolt URI and credentials.
- **minio**: object storage endpoint, credentials, bucket.
- **llm**: OpenAI-compatible base URL, model, token limits, decoding parameters.
- **re**: primary extractor and fallback extractor URLs.
- **csv.thresholds**: biomedical and confidence thresholds used by CSV and graph workflows.
- **classification**: zero-shot model and labels.

5.3 Versioning Strategy

The current FastAPI app declares version 0.1.0, but the REST paths are not versioned by URI or header. Recommended strategy:

- Keep current paths stable as implicit v1 until a breaking change is required.
- Introduce `/v1/...` before external users depend on the API.
- Add only backward-compatible response fields when possible.
- For breaking schema changes, run `/v1` and `/v2` side by side through FastAPI router prefixes or Caddy routing.
- Include `X-API-Version` or an explicit OpenAPI version in release notes for reproducibility.

5.4 Observability

Observed:

- Uvicorn/FastAPI runtime logging through Docker logs.
- Caddy log directive enabled for public ingress.
- `prometheus-client` is listed in API requirements, but no `/metrics` route was observed in mounted routers.
- `monitor_ingestion.sh` polls OpenSearch count/stats and tails API logs for ingestion progress.
- `test_partial_data.sh` probes indexed triplet count and sample search behavior.

Recommended observability additions:

- Request ID middleware and `X-Request-ID` response header.
- Structured JSON logs including method, path, status, latency, tenant, request ID, and upstream service timing.
- Prometheus `/metrics` with request count, latency histogram, error count, queue depth, ingestion throughput, LLM latency, and extraction latency.
- Health/readiness endpoints per dependency: Postgres, OpenSearch, Neo4j, MinIO, Redis, LLM, REBEL, GROBID.
- Alerts for API 5xx rate, search latency, queue backlog, model container restarts, and disk usage in OpenSearch/MinIO/Neo4j volumes.

5.5 Testing Report

Observed test coverage is operational rather than unit-test based. The repository contains `test_partial_data.sh`, which verifies that the triplet index has enough documents and then sends sample search calls for `cancer` and `lung`. No pytest suite, integration test suite, or GUI usability test suite was found in the inspected tree.

Recommended test matrix:

Area	Tests Needed
Security	Missing API key, invalid API key, query-string key, public graph view bypass, missing tenant, cross-tenant access.
Search	Empty query, short query, k limits, backend unavailable, no hits, mixed paper/triplet fields.
Chat	SSE token stream parsing, malformed LLM chunks, citations event, session ID header, long context truncation.
Papers	Unsupported file type, empty file, MinIO unavailable, task helper unavailable, link/PMID payload validation.
Triplets	CSV schema variants, invalid floats, threshold filtering, empty graph IDs, graph viewer with large ID sets.
Annotations	Payload >8KB, invalid cursor, update with no fields, not-found delete, idempotency conflict behavior.
GUI	Login failure, search failure, no selected rows, graph build failure, public API URL generation, history search mismatch.

5.6 Maintenance Risks and Recommendations

- **Mounted-router drift:** Several routers exist but are not mounted. Maintain a route inventory in tests to prevent GUI/API mismatches.
- **Settings compatibility issues:** Some route code references legacy setting names such as `settings.minio_bucket`, while the Pydantic config defines nested `settings.minio.bucket`. Validate these paths under a running container.
- **Security roadmap gap:** HMAC and rate limiting are in config but not implemented in middleware. Either implement them or remove the settings to avoid false assurance.
- **Graph viewer exposure:** `/triplets/graph/view` is public by middleware design. Confirm that graph IDs do not expose sensitive content before public deployment.
- **API versioning:** Add versioned route prefixes before external integration grows.
- **Operational docs:** Add runbooks for indexing recovery, model cache provisioning, backup/restore of volumes, and API key rotation.

6 Appendix A. File and Source Evidence

Primary files inspected:

- `docker-compose.yml`
- `infra/Caddyfile`
- `config/default.yaml`
- `services/api/app/main.py`
- `services/api/app/config.py`
- `services/api/app/middleware/security.py`
- `services/api/app/middleware/tenant.py`
- `services/api/app/routers/search.py`
- `services/api/app/routers/chat.py`
- `services/api/app/routers/papers.py`
- `services/api/app/routers/triplets.py`
- `services/api/app/routers/annotations.py`
- `services/api/app/routers/ingest.py`
- `services/streamlit/app.py`
- `services/api/requirements.txt`
- `services/streamlit/requirements.txt`
- `Makefile`
- `test_partial_data.sh`
- `monitor_ingestion.sh`

7 References and Documentation Basis

This report was generated from the source tree on `blue-demon` at commit `54029058dc360cf000e053d08f6d49979c3bcd63`. The repository URL is <https://github.com/iarroyof/ai-research-insights>. The documentation structure follows common software and API documentation guidance:

- Martínez Cisneros, C. F., Quiroz Bautista, J. U., Guzmán Solano, C. A., García Rivera, B. K., García Pacheco, I., Balderas Martínez, Y. I., Adebayo, K. J., and Arroyo Fernández, I. (2025). *Scalable Construction of a Lung Cancer Knowledge Base: Profiling Semantic Reasoning in LLMs*. 2025 Mexican International Conference on Computer Science (ENC), 1–6. <https://doi.org/10.1109/ENC68268.2025.11311861>
- Balderas-Martínez, Y. I., Sánchez-Rojas, J. A., Téllez-Velázquez, A., Juárez Martínez, F., Cruz-Barbosa, R., Guzmán-Ramírez, E., García-Pacheco, I., and Arroyo-Fernández, I. (2025). *Semantic reasoning using standard attention-based models: An application to chronic disease literature*. *Big Data and Cognitive Computing*, 9(6), 162. MDPI.
- Atlassian software documentation guidance: <https://www.atlassian.com/blog/loom/software-documentation-best-practices>
- Postman REST API best practices: <https://blog.postman.com/rest-api-best-practices/>
- Postman API documentation guidance: <https://www.postman.com/api-platform/api-documentation/>

- Microsoft API design guidance: <https://learn.microsoft.com/en-us/azure/architecture/microservices/design/api-design>
- Microsoft RESTful web API design guidance: <https://learn.microsoft.com/en-us/azure/architecture/best-practices/api-design>